

**Characterizing the Performance of Magneto-Optical Traps by Analyzing the Capture
Velocity of Strontium-88 Atoms through Different Trap Geometries**

Leo Wang

Montgomery Blair High School
Silver Spring, Maryland

Dr. Chad Ropp

National Institute of Standards and Technology (NIST)
Gaithersburg, Maryland

Dr. Stephen Eckel

National Institute of Standards and Technology (NIST)
Gaithersburg, Maryland

Dr. Daniel Barker

National Institute of Standards and Technology (NIST)
Gaithersburg, Maryland

Abstract

The greatest technological innovations that are currently being developed rely on quantum science, ranging from the highest precision measurements to quantum computing. However, these innovations require technologies for working with quantum media, like atoms. There is an apparatus known as the Magneto-Optical Trap (MOT) which laser cools atoms and traps them, allowing them to be used in quantum gates for computation. However, these MOTs are traditionally very cumbersome and can only be operated in a laboratory. This investigation sought to compare several MOT designs and evaluate the performance of an “on-chip” divergent Gaussian beam MOT, a more compact MOT. My project involves scripting these MOT geometries in Python and running a simulation of the MOTs focusing on two main tasks: analyzing the capture velocity of the MOTs and evaluating the performance of a misaligned ‘on-chip’ MOT. After analyzing the results from the scripts, my experiment provided evidence that the ‘on-chip’ MOT performed reasonably given that it featured a miniaturized design. Furthermore, there was evidence that the ‘on-chip’ MOT performance was consistent, even with misalignments. However, the scripts can be refined to be more accurate and these results should be reproduced in a laboratory setting with physical laser beams.

Introduction

The application of the measurement of laser-cooled elements has greatly expanded over the last few decades, ranging from applications in atomic clocks [1] to applications in quantum computing [2]. For example, it is possible to utilize the electron transition frequency of atoms to generate intervals for these atomic clocks. Advancements in the development of atomic clocks have allowed for the accuracy of these clocks to be better than 1 part in 10^{15} [3]. The best way to achieve such accuracy is to use Magneto-Optical Traps (MOTs) because the atoms must be cooled to such a degree where they can produce very precise measurements. Some of the limitations in these clock technologies require advancements in MOT physics. Understanding how elements are able to be cooled in MOTs can lead to advancements in the accuracy and implementation of these various applications. In particular, we analyze the trapping of Strontium-88 atoms in various MOT geometries in order to characterize their effectiveness in capturing atoms.

Magneto-Optical Traps (MOTs)

MOTs have become the modern standard for atomic cooling [1] because they can routinely cool atoms to microkelvin temperatures. Atoms that are cooled are particularly desirable mainly because of the slower speed of their movement, allowing for lower noise measurements. MOTs utilize lasers and magnetic coils in order to cool the atoms by slowing them down, and thus “trapping” them. As seen in figure 1, the MOTs must provide a very

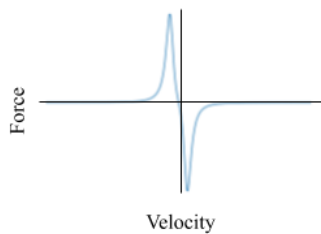


Figure 1: The forces that an atom experiences based on its velocity. If the atom is moving with positive velocity, there needs to be a negative force to push it to the center and vice versa.

specific force profile in order to trap the atom. This force profile must be produced in all three dimensions for the MOT to function. In this setup, the magnetic coils are placed so that they generate a magnetic field while lasers are also incident onto the trap. Six lasers are usually placed at 90-degree orientations so that there is always a force that opposes the motion of the atom. In the system,

the atoms absorb photons from the laser light and lose momentum, thus slowing down. This technique, called Doppler Cooling, utilizes the Doppler

Effect which changes the detuning of the lasers, and the atoms are “dampened” as they become trapped. For example, if the light is detuned red, or at a lower frequency, the atom will absorb more photons if it is moving towards the laser source. It is important to note here that the atoms constantly switch between ground and excited states. After an atom receives a photon, it becomes excited. However, it eventually releases the photon in an arbitrary direction as it drops back to the ground state. Each time the atom switches states, the atom loses momentum from the ejected photon. There are indeed momentum changes that occur here, but the average change in momentum from the emitted photons is 0. The atoms are gradually slowed down through the constant absorption and emission of these photons.

It is important to note that for the purposes of this project, I am analyzing MOTs that are on-chip, rather than conventional six-beam MOTs. Conventional six-beam MOTs utilize six large laser beams that are placed in 90-degree orientations. Since conventional six-beam MOTs are large systems and require table-top optics, they are often impractical for compact commercial applications. On the other hand, on-chip MOTs can be simulated using a beam emitting from one chip. This idea is important because it allows for the creation of a very compact MOT. Many previous experiments utilize very large laser beams (laser spots on the order of inches), which are able to both behave and be modeled as an infinite plane wave [4]. However, since we are trying to model these ‘miniaturized’ MOTs, there will be some discrepancies in the MOT performance. Thus, aside from analyzing the capture potential of the different MOT geometries, we also observe how resistant MOTs are to problems such as azimuthal and radial misalignment of the beams. Research on miniaturizing MOTs is important because it allows for more practical use in accelerometers and clocks: traditional MOTs require large amounts of equipment in a lab setting to function.

Gaussian Beams

In an experimental setting, it is impossible to create infinite plane waves, which on simple terms is a laser beam with infinite width and uniform intensity in all space. In order to take into account this constraint, we aim for lasers that can follow a Gaussian beam profile. Figures 2-4 show the Gaussian beam’s intensity profiles from different slices in the $z=0$ plane. It is easier to characterize this type of beam because there are already predefined equations that would allow us to describe the lasers analytically.

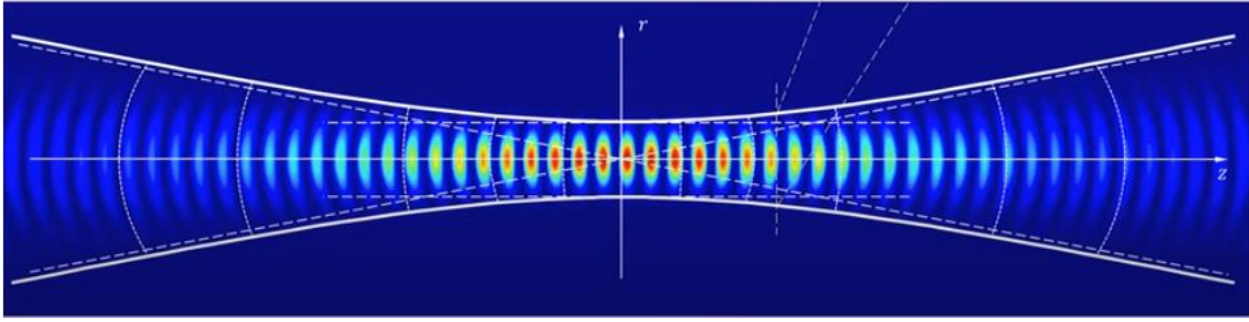
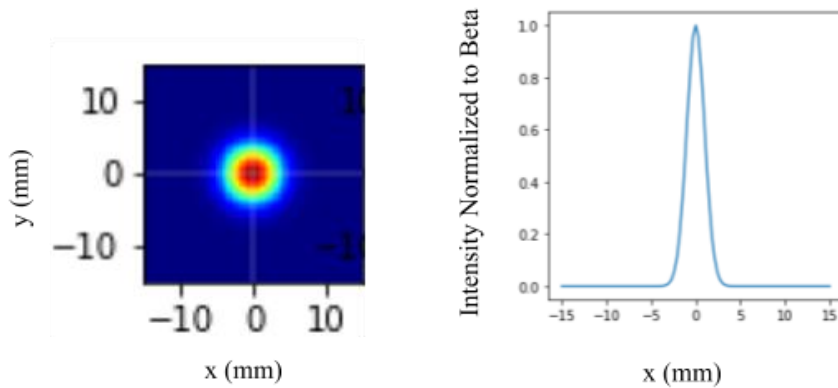


Figure 2: Gaussian intensity profile. Retrieved from Rwthachen University.



Figures 3-4: Gaussian intensity profiles from different perspectives of the beam.

Related Experiments

The idea to miniaturize the conventional six-beam MOT with a chip and gratings is not new. In 2019, Barker et al. demonstrated a single-beam slowing and trapping apparatus for Li atoms [5]. By using a nanofabricated diffraction grating, they were able to generate a MOT using just one input beam, rather than the conventional six. This allowed the laser beam to diffract upon hitting the grating, creating several beams, allowing for a MOT. Using this methodology, they were able to trap 10^6 Li atoms. This relates to my experiment because one goal of my research is to develop a compact MOT. We can capture Strontium-88 atoms and taking note of

the chip placement of the trap, we will systematically misalign the lasers to test the performance of the trap.

Another experiment by Roy et al. in 2018 looked at some of the applications of misaligned MOTs [6]. They utilized two different misalignment geometries: a “vortex-MOT” and a “hybrid-MOT” where they used both Zeeman-shifted components and optical scattering force components. Using the MOTs, they were able to trap 6×10^6 and 26×10^6 atoms of Rubidium-85 in the vortex-MOT and hybrid-MOT, respectively. Through the analysis of the temperatures of the MOTs, they showed that these MOTs met the temperature threshold to work with quantum sensors and other applications. In my experiment, we will misalign the lasers azimuthally and radially in various increments. This will allow us to observe the capture performance of the MOT at a particular geometry. Aside from testing how robust these traps are, I will also analyze the capture velocities at each specific increment to find a peak.

Rationale

There are several reasons why this research is being completed. First, I aim to compare an on-chip divergent MOT geometry with a conventional six-beam MOT. This will determine the effectiveness of the different MOT types. Second, our research helps characterize azimuthal and radial misalignments of MOT geometries. By looking at very small increments of misalignments, I will be able to determine the peak capture velocity of the MOT. This is important because a more efficient MOT is useful for practical applications such as accelerometers and atomic clocks [7]. Furthermore, my research characterizes the performance of Strontium-88 in MOTs, which will be useful for future research using this element. In creating a script to simulate all of these MOTs, I will not only research the properties of the MOTs but also create a useful tool that can be used to simulate future MOTs. This tool will be particularly useful for those who are not able to simulate atoms in lab conditions.

The main way we plan to simulate the MOTs is to program a Python script to characterize the lasers and atoms. This script will then use the well-defined mathematical calculations to simulate a MOT in specified conditions.

Methods

Scripting

Due to the COVID-19 pandemic, we were not able to access the physical lasers to perform experiments, rather we chose to script our simulations. Using some starting code from the lab's 'pylcp' project where they ran other MOT simulations, I created a separate branch in GitHub and conducted most of my research there. To run our scripts and generate the graphs, I used the Anaconda distribution of Python along with JupyterLab for easy script management.

Heuristic Equation

Since physical MOTs were not available, the first step in our research was to develop a robust way to simulate the MOT. To simulate the background force profile of our MOT, we used a heuristic equation that provided a good model for forces present in the MOT. Although this equation was not perfect, it was accurate enough for the purposes of this experiment. Since the experiment dealt with Strontium-88, there were only two states we had to worry about, the excited and ground states. Taking these states into account, we used the following equation as a model of the forces in our MOT:

(1)

$$F = \frac{\hbar k \Gamma}{2} \sum_{q,i} \frac{\beta_i (\epsilon_q \cdot \hat{B}(r))^2}{1 + \sum_j \beta_j + 4[\Delta - \mathbf{k}_i \cdot \mathbf{v} - q\mu_B B(r)/\hbar]^2/\Gamma^2}$$

In this heuristic equation, F represents the force, Γ is the decay rate of the excited state, $q = -1, 0, 1$, which is the spherical basis, $\beta_i = I/I_{sat}$ where I is the intensity of the laser and $I_{sat} = \hbar\omega^3\Gamma/(12\pi c^2)$ is saturation intensity for an equivalent two-level atom where ω is equal to the frequency multiplied by a factor of 2π , v is the velocity, ϵ_q is the q th polarization component, and k_i represents the intensity and wavevector for the i th beam, which is based on time t and position r . It is also important to remember the constants, $\hbar$, the reduced Planck's constant, and μ_B , the Bohr magneton in the equation. Since I was simulating this MOT in a 3D quadrupole field, we used the following equation for B , the magnetic field:

(2)

$$\mathbf{B} = B' \left(-\frac{1}{2}(x\hat{x} + y\hat{y}) + z\hat{z} \right)$$

In this equation, B' is set to be the magnetic field gradient, which is the linear variation in the magnetic field from one position to another position. We also set $\hat{x}, \hat{y}, \hat{z}$ as the unit vectors for the direction and x, y, z as the coordinates for the system. These variables are then multiplied by a factor of $-\frac{1}{2}$ to satisfy Maxwell's equations. The heuristic equation could then be simplified for the purposes of this project. By choosing to calculate everything from the perspective of the z -axis, we could set a more simplified version of the equation for the force:

(3)

$$\bar{\mathbf{F}} = \frac{\hat{\mathbf{k}}}{2} \sum_{q,i} \frac{\beta_i (\epsilon_q \cdot \hat{B}(r))^2}{1 + \sum_j \beta_j + 4(\delta - \hat{\mathbf{k}}_i \cdot \bar{\mathbf{v}} - q\bar{z})^2}$$

For these equations, we used arbitrary units for the force, velocities, and position. Allowing $\delta = \Delta/\Gamma$ to be the normalized laser detuning, we can determine the velocities in terms of Γ/k , position in terms of $\hbar\Gamma/\mu_B B'$, force in terms of $\hbar k\Gamma$, and time in terms of kx_0/Γ .

Setting up Units

It is important to note that I could always convert these units to SI units by carrying out a specific operation (multiplying and dividing by constants) on the current values. With this in mind, I continued by implementing certain details relevant to the MOT (Table 1) in the script. These details would later be referenced in the script for calculations.

Table 1: This table defines some of the typical units and values relevant to our MOT.

Values Relevant to the Experiment			
Value	Significance	Units	Value
k	Wavenumber	cm^{-1}	$2\pi/\lambda$
d	Position source of atom	mm	Varies based on MOT
Γ	Decay rate	s^{-1}	$2\pi \times (30\text{MHz})$
$t0$	Time unit	μs	10.0
$x0$	Length unit	mm	1.0
β	Intensity relative to saturation	Normalized units	1.0
I_{sat}	Saturation Intensity	mW/cm^2	40.0
wb	Laser beam waist	mm	4.7
Δ	Detuning frequency	MHz	-45.0

Setting up the MOTs

The first goal of the project was to compare the performance of the on-chip divergent MOT with the conventional six-beam MOT. To do this, I first needed to implement these different MOT geometries into the script. Being part of the larger ‘pylcp’ project at NIST, the conventional-six beam MOT was already defined in the package. This allowed me to import it and use it as a starting point for the other MOT configurations. The following code was used to define the k-vectors, the wave-vectors indicating the direction of propagation, and polarization, the direction in which the electric fields of the light points, of the laser beams in the conventional-six beam MOT class:

(1) k-vectors

```

kvecs = [np.array([ 1.,  0.,  0.]), np.array([-1.,  0.,  0.]),
          np.array([ 0.,  1.,  0.]), np.array([ 0., -1.,  0.]),
          np.array([ 0.,  0.,  1.]), np.array([ 0.,  0., -1.])]
pols = [-pol, -pol, -pol, -pol, +pol, +pol]

```

The k-vectors were defined using NumPy, which allows for easier matrix operations through Python. For the conventional six-beam MOT, there were only two polarizations possible for the lasers, positive and negative. After importing this into Python, the next step was the actual implementation of the on-chip divergent MOT. I defined respective classes in the script for the implementation of this MOT. For both MOT implementations, I used a consistent set of MOT characteristics listed above (Table 1). However, I also needed to include circular polarization for the on-chip divergent MOT (specific to this MOT):

(2) Circular polarization

```

self.add_laser(pylcp.gaussianBeam(kvec=kmag*np.array([0., 0., -1.]), \
    pol=np.array([1., -1.j, 0])/np.sqrt(2), beta=1.0*beta, delta=delta, \
    wb=4.65, r0=np.array([0., 0., 0.]), **kwargs))
self.add_laser(pylcp.gaussianBeam(kvec=kmag*np.array([0.612, 0.75, -0.25]), \
    pol=np.array([1., 1.j, 0])/np.sqrt(2), beta=1.0*beta, delta=delta, \
    wb=4.65, r0=np.array([0., 0., 0.]), **kwargs))
self.add_laser(pylcp.gaussianBeam(kvec=kmag*np.array([-0.612, 0.75, -0.25]), \
    pol=np.array([1., 1.j, 0])/np.sqrt(2), beta=1.0*beta, delta=delta, \
    wb=4.65, r0=np.array([0., 0., 0.]), **kwargs))
self.add_laser(pylcp.gaussianBeam(kvec=kmag*np.array([0., 0., 1.]), \
    pol=np.array([1., -1.j, 0])/np.sqrt(2), beta=6.261e7*beta, delta=delta, \
    wb=5.744e-4, r0=np.array([0., 0., -17.788]), **kwargs))
self.add_laser(pylcp.gaussianBeam(kvec=kmag*np.array([-0.612, -0.75, 0.25]), \
    pol=np.array([1., 1.j, 0])/np.sqrt(2), beta=6.261e7*beta, delta=delta, \
    wb=5.744e-4, r0=np.array([10.893, 13.341, -4.447]), **kwargs))
self.add_laser(pylcp.gaussianBeam(kvec=kmag*np.array([0.612, -0.75, 0.25]), \
    pol=np.array([1., 1.j, 0])/np.sqrt(2), beta=6.261e7*beta, delta=delta, \
    wb=5.744e-4, r0=np.array([-10.893, 13.341, -4.447]), **kwargs))

```

After setting up instances of these classes, the background force profile of the MOTs was calculated. I then simulated the capture of atoms at different initial velocities and generated relevant graphs. To do this, I set a capture and lost condition for the traps:

(3) Captured and lost conditions

```

def captured_condition(t, y, threshold=1e-2):
    if np.linalg.norm(y[-6:-3]) < threshold:

def lost_condition(t, y):
    if np.linalg.norm(y[-3:]) > 15.:

```

The captured condition allowed me to determine if the atom was at a near-zero speed near the center of the MOT. The lost condition allowed me to test if the atom had reached a certain distance so that I could be reasonably sure it had escaped the trap (15 mm in this case). I then released atoms at speeds from 10 to 90 m/s into the MOT and generated relevant graphs. To provide a deeper analysis of the MOTs, I chose to run the model under different β and δ s while recording the capture velocity. To do this, I utilized the following Python method:

(4) *Capture velocity*

```
def bisectFindChangeValue(fun, initial_guess, args=(), kwargs={},
                          maxiter=1000, tol=1e-9, invert=False, debug=False):
```

This method takes in the initial positions, the heuristic equation, and the capture and lost conditions and uses them to find a capture velocity (maximum velocity before the MOT can no longer capture the atom; all slower atoms will be assumed to be captured) of the atom when it is trapped. I ran this method with the following set of β s and δ s for each MOT:

(5) *β and δ*

```
dets = gammabar*np.linspace(-3, -0.5, 41)
betas = np.array([0.5, 1., 1.5])
```

I then graphed the capture velocity as a function of β and δ for each of the MOTs for comparison.

Misaligning the MOTs

To test how robust the on-chip divergent MOT is, I systematically misaligned the lasers in two ways: azimuthally, where the beams shift and miss one another, and radially, where the beams still hit one another, but at a shifted height (Figures 5-6). To implement this into the script, I chose to use comma-separated (CSV) files that include incremental shifts of the beams. We created one CSV file for each argument that is needed to define the MOT and populated each file with 13 rows and 15 columns. Each cell of the CSV file represents a particular misalignment with these misalignments becoming incrementally increased towards the outer rows and columns. It is important to note that the center cell for all of the files is the on-chip divergent MOT that I simulated in the previous section, where the geometry of the beams is perfectly aligned. I then used for loops to run each misaligned MOT in the `bisectFindChangeValue()`

function and stored the capture velocities in a new CSV file, which is saved in the file directory. I then wrote the following code to generate a colormap of the capture velocities for each MOT using Matplotlib:

(6) *Colormap of normalized capture velocities*

```
for r in range(len(dataOfVc)):
    for c in range(len(dataOfVc[r])):
        dataOfVc[r][c] = dataOfVc[r][c]/normalize

plt.imshow(dataOfVc, extent=[-30,30,-35,35], origin='bottom')
```

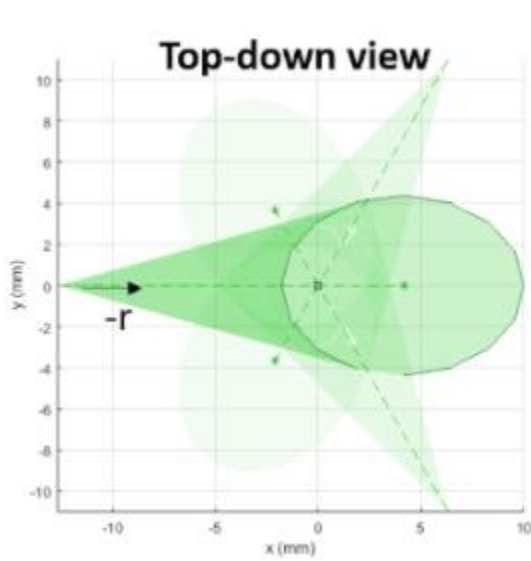


Figure 5: The top-view of a radial laser misalignment.

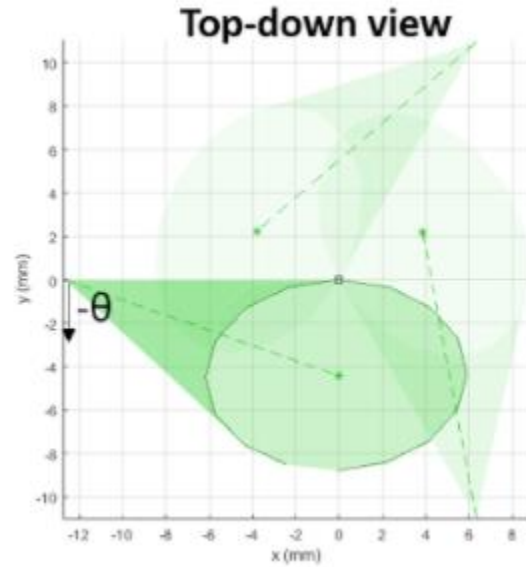


Figure 6: The top-down view of an azimuthal laser misalignment.

Results

MOT Performance

In the conventional six-beam MOT, Strontium-88 atoms with speeds from up to 60 m/s were successfully cooled and trapped. This can be seen in the phase diagram in Figure 7, where

the first six atoms released from position $(0, 0, -10)$ were trapped in the center of the graph, where they evolved towards a state with a near-zero velocity at the center, $z=0$ position. In the figure, each white line corresponds to an atom's trajectory; atoms with initial velocities ranging from 10 to 100 m/s were simulated (seen on the y-axis). In the phase diagram, the atom's trajectory is summarized from the perspective of the z -axis, where the color of the diagram represents the force experienced by the atom. The velocity of the atom changes based on the force experienced from the laser at a given position. For example, for the atom with an initial speed of 10 m/s, it is initially sped up by the force but eventually slowed down until it is captured at the center $(0, 0, 0)$.

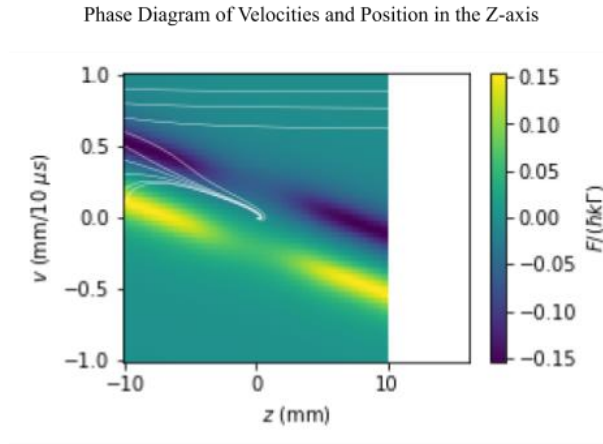


Figure 7: The phase diagram for the conventional six-beam MOT, with each white line representing a single trajectory.

However, this phase diagram graph could not be replicated for the on-chip divergent MOT because of the difference in the atom starting location. Where I can simulate the conventional six-beam MOT from only the z -axis, this is not possible for the on-chip divergent MOT which has a starting location of $(0, -8.839, -8.839)$. As a result, I cannot compare the two MOTs by looking at the trajectories. Therefore, I generated the magnitudes of the position and velocities for the trajectories of the atoms through both MOTs, which can be compared (Figures 8-11).

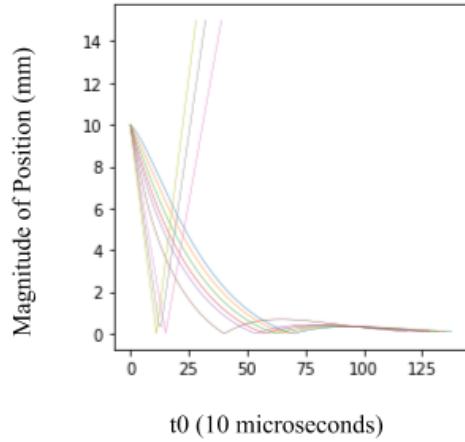


Figure 8: The magnitude of the position of the atoms through the conventional six-beam MOT.

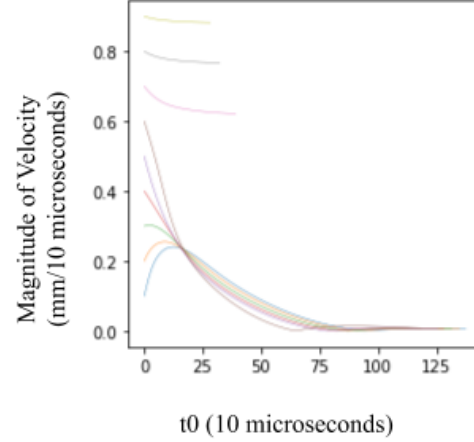


Figure 9: The magnitude of the velocity of the atoms through the conventional six-beam MOT.

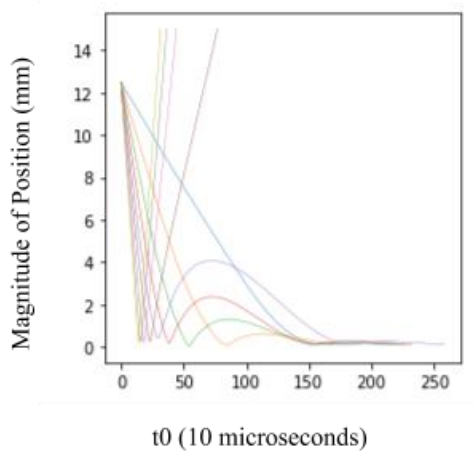


Figure 10: The magnitude of the position of the atoms through the on-chip divergent MOT.

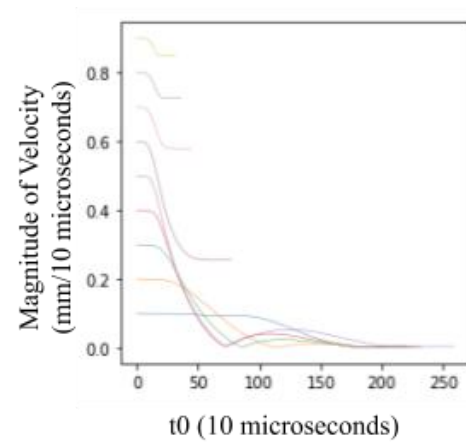


Figure 11: The magnitude of the velocity of the atoms through the on-chip divergent MOT.

In the conventional six-beam MOT, Strontium-88 atoms with speeds up to 60 m/s were successfully cooled and trapped. This can be seen in Figures 8-9, where the first six atoms released from (0, 0, -10) quickly went to near-zero velocity and position, whereas the other atoms (speeds 70 to 90 m/s) continued traveling away from the MOT. It is important to note that our script stops recording the position and velocity of a lost atom once the lost condition has

been satisfied. At this point, the atom escapes the system and moves off at a constant velocity. Otherwise, the velocity goes to 0 and the capture condition is satisfied. In the on-chip divergent MOT, Strontium-88 atoms with speeds from 10 to 50 m/s were successfully cooled and trapped. This can be seen in Figures 10-11, where the first five atoms released from (0, -8.839, -8.839) reached near-zero velocity and position, whereas the other atoms (speeds 60 to 90 m/s) continued traveling away from the MOT.

I also plotted the maximum capture velocities of the on-chip divergent and conventional six-beam MOTs (Figure 12). I generated the graph with β values from 0.5 to 1.5 and 41 different δ s from -3.0 to 0.5 for both MOTs, which are distinguished by line type. If I only restrict a β value of 1.5, the conventional MOT peaks at 68.1 m/s (rounded to one decimal), and the on-chip divergent MOT peaks at 58.2 m/s (-14.5% difference). For a β value of 1.0, the conventional MOT peaks at 64.3 m/s, and the on-chip divergent MOT peaks at 55.5 m/s (-13.7% difference). For a β value of 0.5, the conventional MOT peaks at 56.7 m/s, and the on-chip divergent MOT peaks at 49.1 m/s (-13.4% difference).

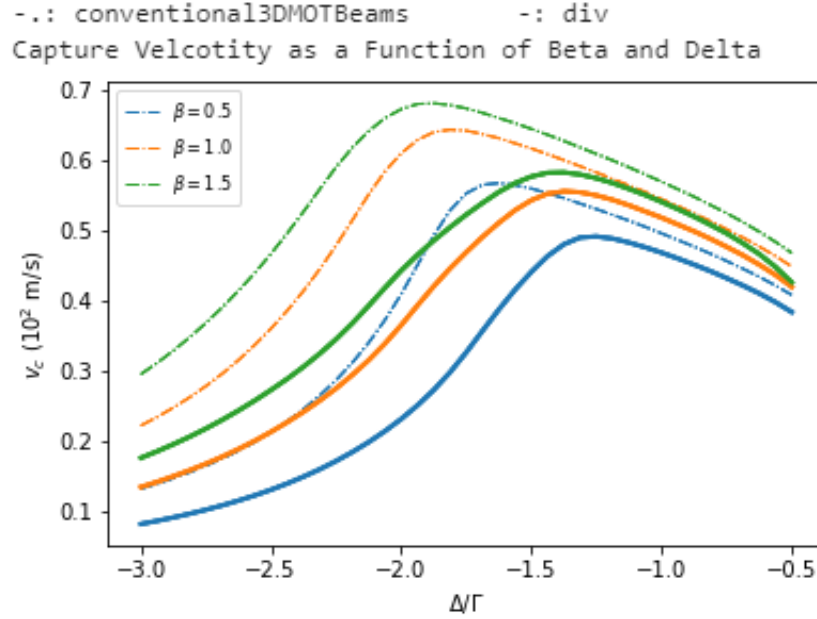


Figure 12: This graph displays the change in capture velocity for a given beta and delta for the on-chip divergent and conventional six-beam MOTs.

Misalignments

After generating the capture velocities for each of the 195 misaligned versions of the on-chip divergent MOT, I used these values to create a color map (Figure 13). Instead of plotting just the capture velocity, I plotted the capture velocity raised to the fourth power normalized to 1 because this value, v^4 , has the property where $v^4 \propto n$, where n is the total number of atoms trapped in the MOT [8]. Using the colormap, we notice that the peak v^4 is not actually in the dead center with no misalignments. Instead, I find that peak v^4 occurs when the MOT is radially misaligned (dr) by around 10 μm .

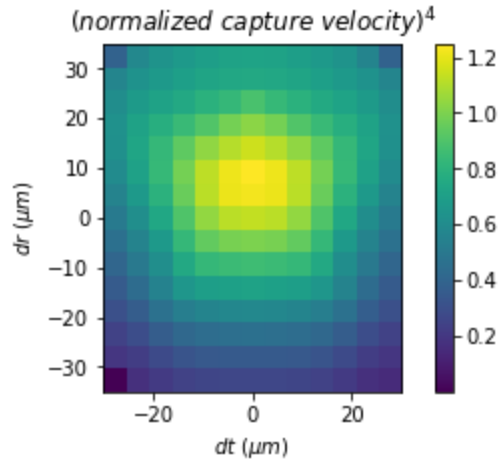


Figure 13: A colormap based on the capture velocities calculated from the misalignment of the on-chip divergent MOT.

Discussion

Capture Velocity

One of the goals of this project was to analyze the performance of the on-chip divergent MOT that is being developed by NIST with the conventional six-beam MOT. Through the velocity and position magnitude graphs, I found that although the conventional six-beam MOT performed the best, being able to capture the atom traveling at 60 m/s, the on-chip divergent MOT was not far behind, successfully capturing the 50 m/s atom. This is significant to our project because it shows that although the on-chip divergent MOT is miniaturized for better applicability, its performance is close to that of the conventional six-beam MOT, which often requires a lab setting.

These findings are also supported in the combined capture velocity graph that is based upon different values of β and δ (Figure 12). In analyzing the graph, we note that the number of atoms trapped is directly proportional to the capture velocity raised to the fourth power (v^4). This means that by looking at the capture velocities, we can get a general sense of the capture potential of the different MOTs. We see that although the conventional six-beam MOT outperformed the on-chip divergent MOTs by a minor margin (13.4% to 14.5% increase in capture velocity when compared to on-chip), the on-chip divergent MOTs were still able to

produce generally effective trapping given that they were miniaturized. From the graphs, we also see that there is one clear peak for both MOTs. Starting at -3.0δ , both MOTs increase in performance until around -2.0 to -1.5 , where they peak and quickly drop off. This is generally fine because any of these δ s and β s for the conventional MOT can be simulated in lab conditions. However, this finding is important because, for the on-chip divergent MOT, we can only change the δ , but not the β due to power constraints. Larger β s are usually associated with improvements in MOT performance, but we cannot make it arbitrarily large due to the aforementioned limitations. Thus, we must test different values of β experimentally to see the trend.

There are several ways this section of the simulations can be improved to alleviate some possible sources of error. Firstly, if we are to repeat this experiment, we can fine-tune some of the functions used to calculate the atom's trajectory. In the scripts, we mostly relied on integrators which would integrate over a specific time interval and return the atom's trajectory. If we were to integrate at shorter time intervals, we would get a better model for the trajectory. We could apply the same principle to the velocity. For example, we can simulate more atoms at smaller increments for the initial velocities. Likewise, we can also alter the `bisectFindChangeValue()` function and the capture conditions by making the threshold tighter. This way, we can obtain a more accurate value for when the atoms are captured. Also, by setting more values for β and δ , we can do a better job of pinpointing where exactly the peaks are for the capture velocities. However, it should be noted that if these changes were made, the models would take significantly longer to run. Also, although the heuristic equations provided a *good* model, there are other models that are more accurate, such as the rate equation model. As one progresses from the heuristic equation, to the rate equation, and to the optical bloch equation models, the complexity and precision increases. The heuristic model assumes just a two-level atom, which works well for Strontium-88. Although the rate equation model would work better for Strontium-88, going to the optical bloch equations is not necessary since Strontium-88 is a relatively simple two-level atom. For an atom such as Strontium-87, going to the optical bloch equations would be beneficial due to the more complex level structure. During this experiment, we did deploy a script that made the same capture velocity graphs with the rate equations, but we were not able to simulate all of the MOTs simply because the rate equations are calculation-

intensive and require a very long computing time (on the order of days). Thus, we made the decision to not simulate the rate equations and chose fair thresholds for our calculations.

Misalignments

There are two impactful conclusions from the misalignment section of our experiment. Looking at the data from the 195 (13 x 15) misalignments of the on-chip divergent MOT (Figure 13), we see that the MOT is very robust. From the colormap, we see that for azimuthal misalignments (dt) as severe as $\pm 25 \mu\text{m}$, v^4 is relatively stable, and the MOT captures roughly 50% of the atoms that can be captured. The radial misalignments (dr) did not seem to impact the MOT as much, with the $\sim 40\%$ decrease in v^4 only starting to occur at $+30 \mu\text{m}$ and $-20 \mu\text{m}$. All of these misalignments are severe, so it is surprising that they do not greatly affect the MOTs. Although there seemingly is a large percent difference, the key is that the v^4 s are maintained on the same order of magnitude. Next, we also see that if we misalign the MOT carefully, we are actually able to improve our result, allowing us to capture more atoms. From the colormap, we see that the peak occurs when the MOT is radially misaligned by around $10 \mu\text{m}$. This observation is particularly useful because it means that we can intentionally misalign our Gaussian beams if we want to increase the capture potential for better applicability.

There are a few ways that this part of the experiment can be improved. For one, we used heuristic equations, so if one were to want a more accurate idea of when the capture velocity changes occur, they could utilize a model like the rate equations. Similarly, the model used a crude approximation for saturation, which can be improved by changing our heuristic model. Furthermore, like the previous section of our experiment, we can also decrease misalignment increments so we can have a more accurate idea of the peaks/changes occurring.

Further Experiments

Aside from running simulations under more accurate models, there are a few immediate follow-up projects that can occur. Firstly, one can compare the results our model generated with experimentally obtained values from the lab. Aside from verifying general trends in the data, one can also be more precise and simulate a wider range of values in the lab (such as more velocities, β s, or δ s). For example, it is possible to purposely misalign the laser beams in the lab and determine if certain misalignments improve the capture potential of the MOTs.

A more radical approach for an experiment would be to change the four-state level atom model we simulated with a more complex system, such as the forty-state level atom model with an element such as Strontium-87. This would allow for more comparisons in determining which elements are more favorable for use in MOTs. If these future experiments were to take place in a lab, we could not only verify but expand on current results.

Literature Cited

- [1] Koller, S., Grotti, J., Vogt, S., Al-Masoudi, A., Dörscher, S., Häfner, S., Sterr, U. and Lisdat, C., 2017. Transportable Optical Lattice Clock with 7×10^{-17} Uncertainty. *Physical Review Letters*, 118(7).
- [2] Fairbank, W., Lee, S., Czajkowski, W., Kluck, J., Ralph, T. and Lam, P., 2011. A Laser-cooled Single-atom-on-demand Source For Si Quantum Computing.
- [3] Kitching, J., Knappe, S. and Donley, E., 2011. Atomic Sensors – A Review. *IEEE Sensors Journal*, 11(9), pp.1749-1758.
- [4] Lin, Y., Perry, A., Compton, R., Spielman, I. and Porto, J., 2009. Rapid production of R87b Bose-Einstein condensates in a combined magnetic and optical potential. *Physical Review A*, 79(6).
- [5] Barker, D., Norrgard, E., Klimov, N., Fedchak, J., Scherschligt, J. and Eckel, S., 2019. Single-Beam Zeeman Slower and Magneto-Optical Trap Using a Nanofabricated Grating. *Physical Review Applied*, 11(6).
- [6] Roy, R., Rushton, J., Dragomir, A., Aldous, M. and Himsworth, M., 2018. A misaligned magneto-optical trap to enable miniaturized atom chip systems. *Scientific Reports*, 8(1).
- [7] Bonnin, A., Zahzam, N., Bidel, Y. and Bresson, A., 2013. Simultaneous dual-species matter-wave accelerometer. *Physical Review A*, 88(4).
- [8] Monroe, C., Swann, W., Robinson, H. and Wieman, C., 1990. Very cold trapped atoms in a vapor cell. *Physical Review Letters*, 65(13), pp.1571-1574.